



Indian Institute of Science Education and Research (IISER)  
Mohali

Sector 81, Knowledge City, S A S Nagar, Mohali, Punjab-140 036

B

CHM 102 Basic Organic Chemistry  
End-semester Exam – Spring 2024

Date: 23.04.2024

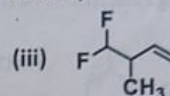
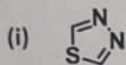
Time: 3 Hrs.

Max Marks: 50

Answer ALL questions

**IMPORTANT:** Each one of the questions "Q1 to Q5" need to be started from a fresh page.

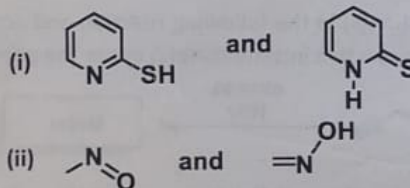
Q1. (a) Write the correct IUPAC names of the following compounds: (5 marks)



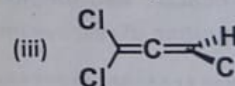
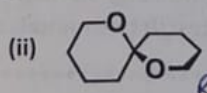
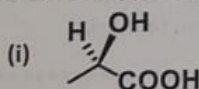
(b) Which among cyclohexanol and phenol is more acidic? Indicate the reason. (2 marks)

(c) The dipole moments of formaldehyde (HCHO) and carbon monoxide (CO) are 2.43 and 0.11 D, respectively. Why? (2 marks)

(d) Identify the relationships between the following pair of molecules. (1 mark)

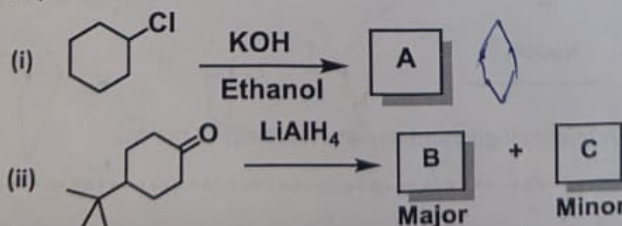


Q2. (a) Redraw the structures, and indicate the appropriate stereochemical notation(s) (among E/Z, R/S and P/M pairs) for the following molecules. Also, identify the type of isomerism/chirality, whichever is applicable. In case if the molecule is achiral, indicate the reason. (3 marks)

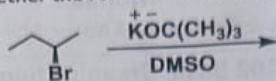


(b) Draw gauche- and anti- conformations of 2,3-dimethylbutane (tetramethylethane). Identify the major conformer among them, and indicate the reason for it. (3 marks)

(c) Predict the major/minor products (if applicable) and draw the most stable conformer(s) of the products. (4 marks)

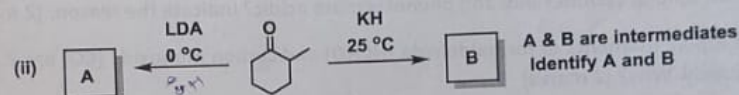
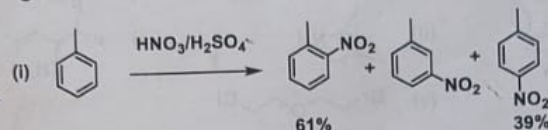


- Q3. (a) Predict the major and minor product(s) in the following reaction along with arrow-push mechanism. Also, identify whether the reaction is entropy driven or enthalpy driven? (3 marks)

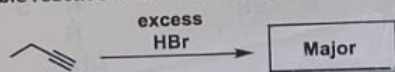


- (b) Using energy profile diagram, explain why radical bromination is selective than radical chlorination in branched alkanes? Note that the rate of radical chlorination is faster than the radical bromination. (4 marks)

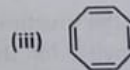
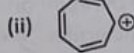
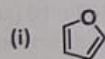
- (c) Construct energy profile diagrams and explain the product/intermediate selectivity in the following reactions. (4 marks)



- Q4. (a) Predict the major product(s) in the following reaction and account for its formation based on the stability of all possible reactive intermediate(s) under the given condition. (3 marks)

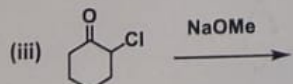
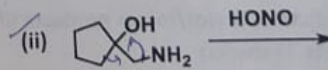
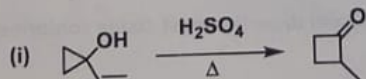


- (b) Classify whether the following molecule is aromatic/anti-aromatic/non-aromatic/homoaromatic/heteroaromatic? (3 marks)



- (c) Indicate two methods each (including starting materials, reagents and conditions) for the generation of reactive intermediates: (i) free radicals and (ii) carbene. (4 marks)

- Q5. (a) Predict the product(s) / arrow-push mechanism / reactive intermediate(s) involved in the following reactions, whichever is/are appropriate. (2 + 3 + 3 marks)



- (b) How to differentiate singlet and triplet carbenes? (2 marks)

Bonus question:

Explain the reason for the product selectivity in

